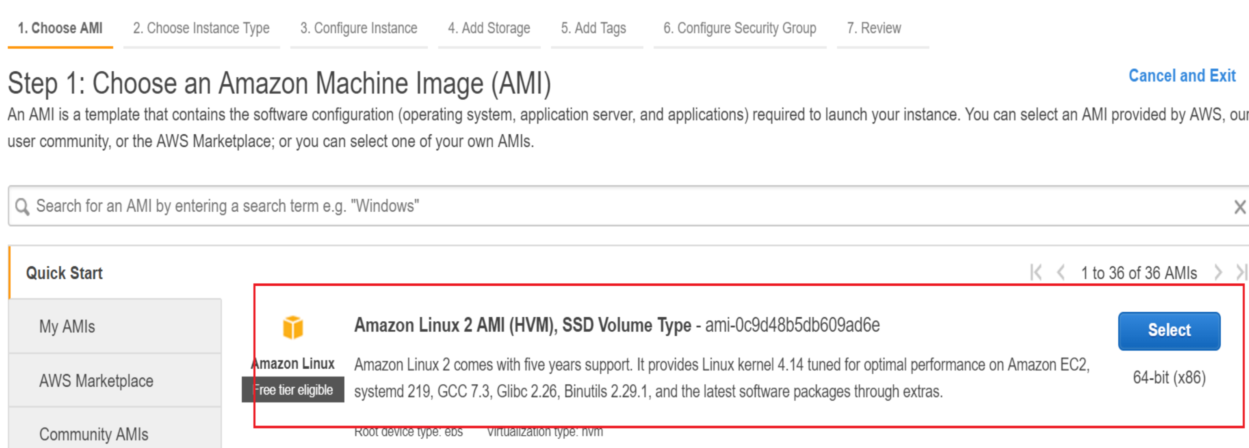
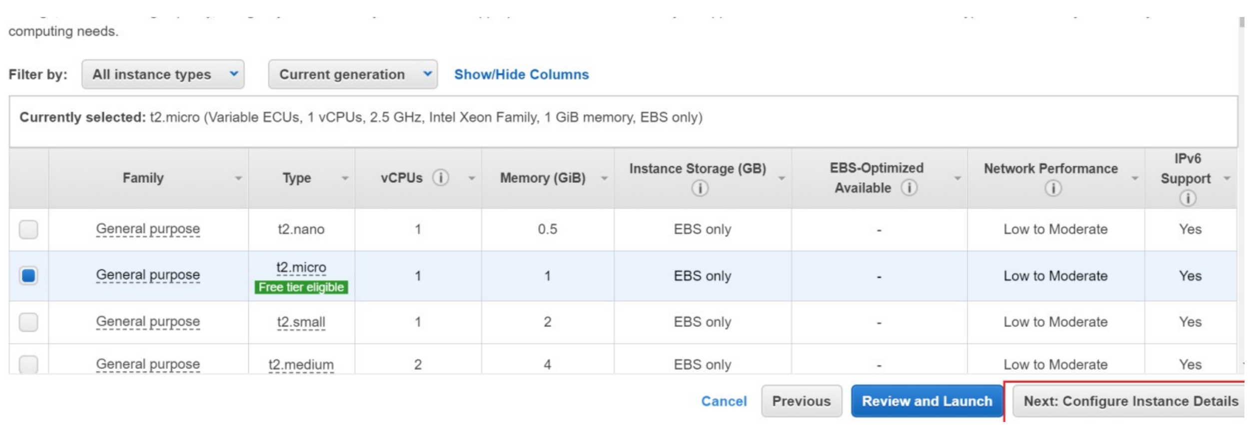
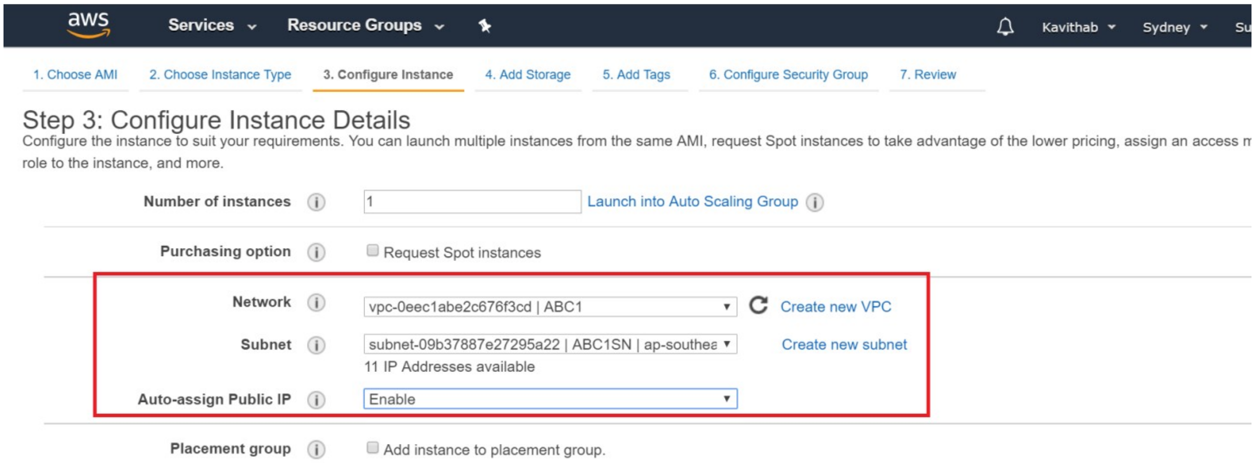
Cloud Watch Steup

Step1: -





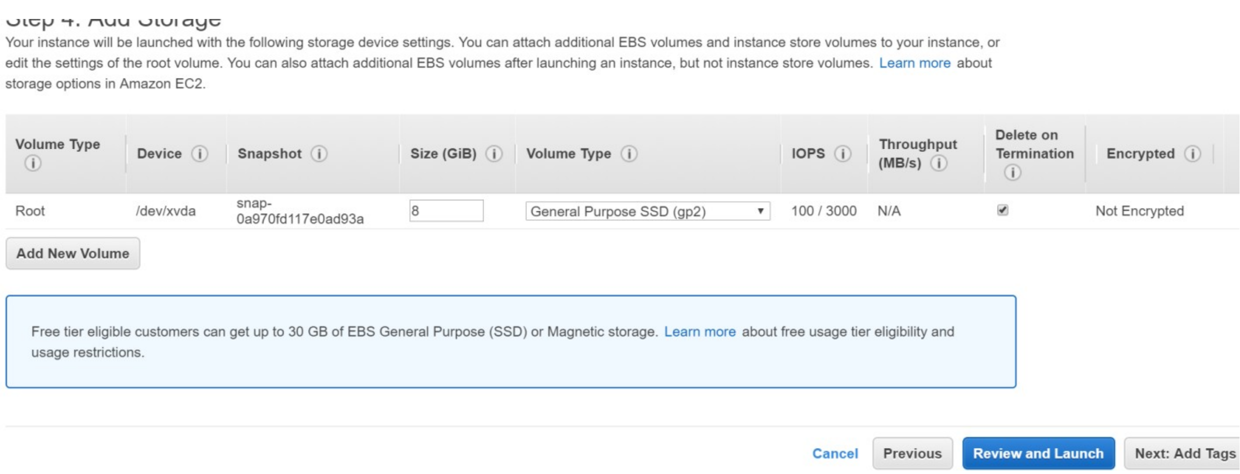
Click on Configure instance details



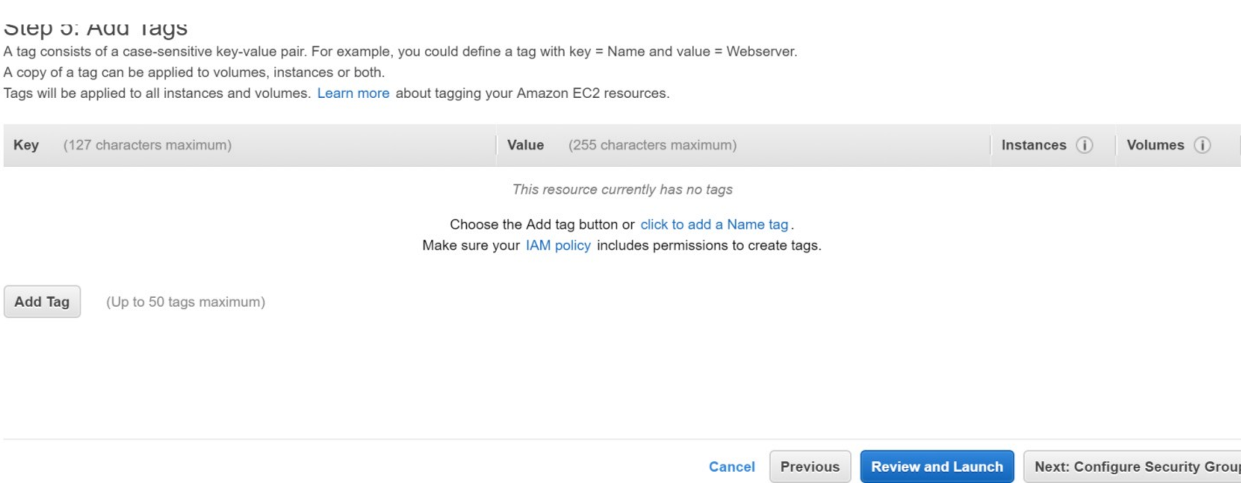
Enter the details of Network (VPC)

Subnet (auto selection) and enable the Public IP as in the above screen shot.

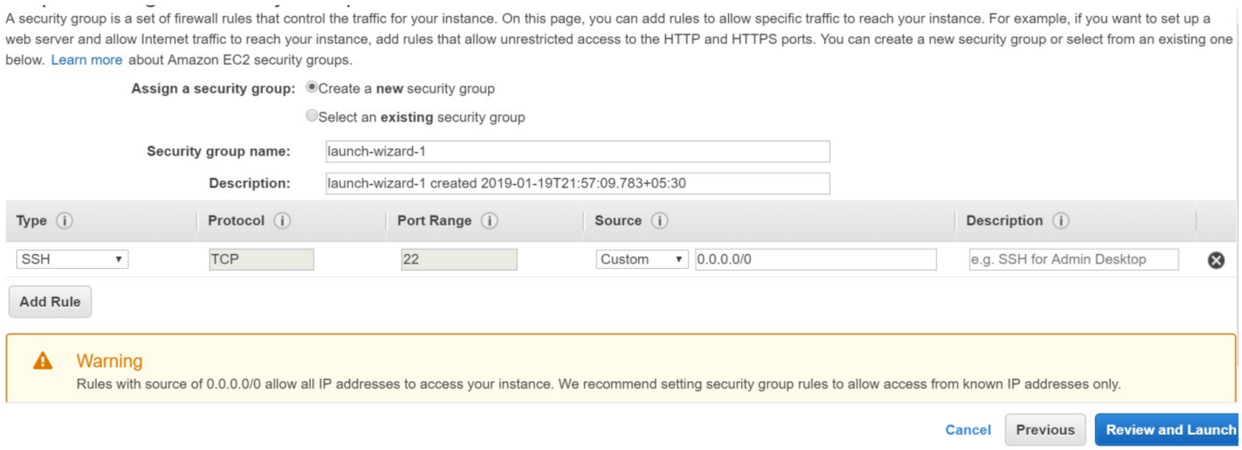
Then click on add storage



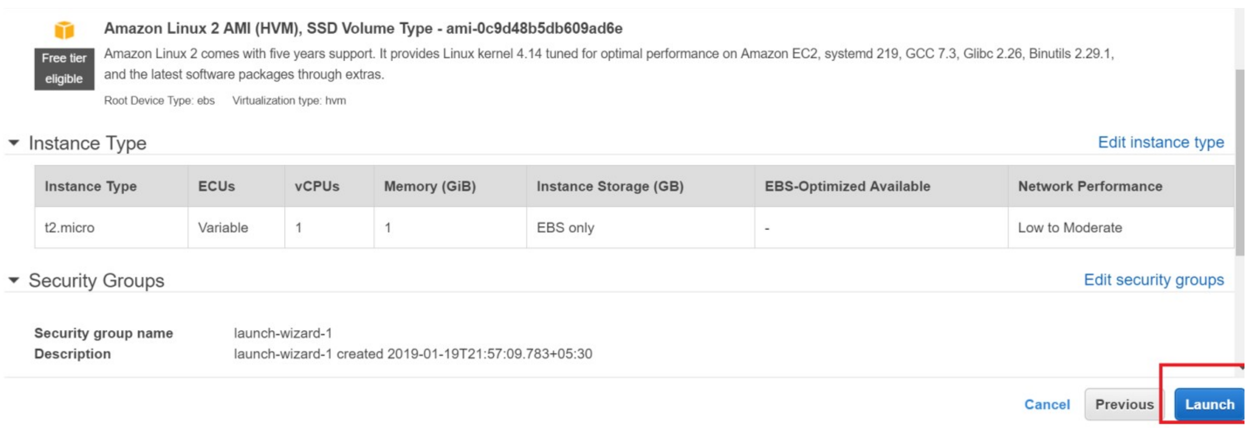
Click on Add tag (nothing to be enter in the above screen)



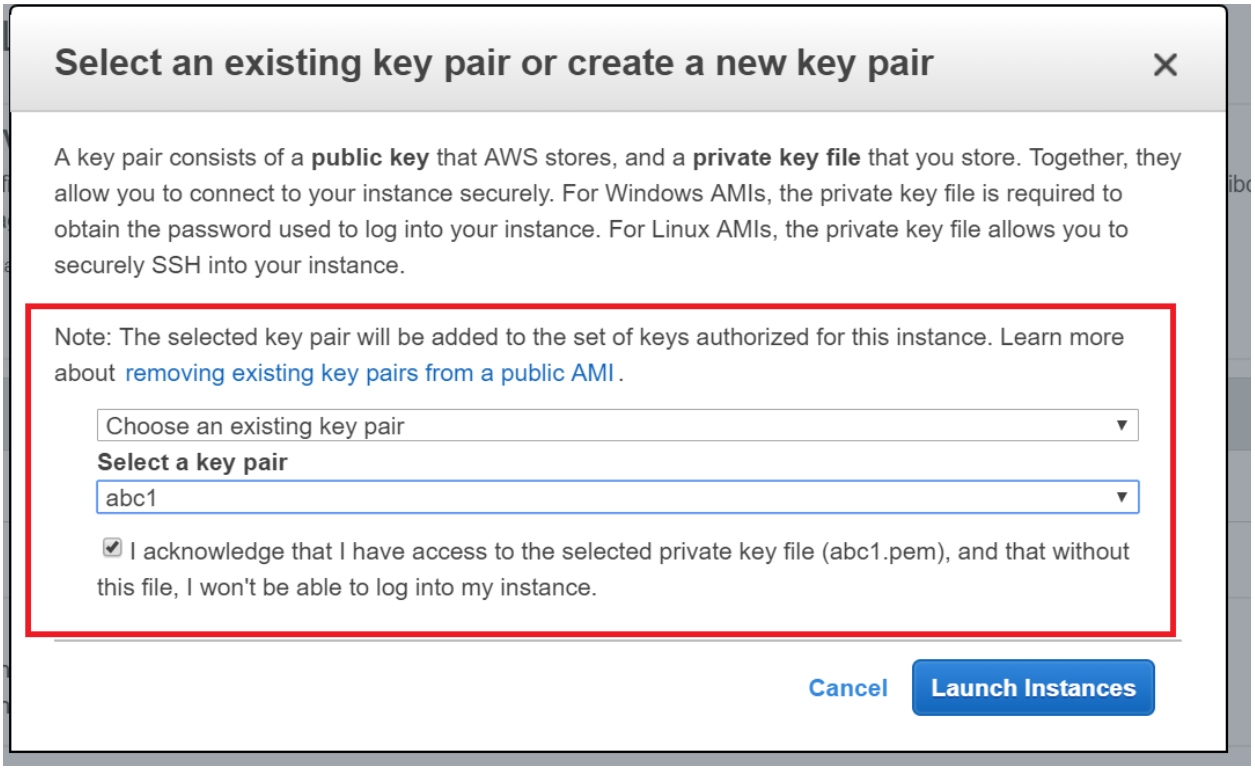
Click on Configure Security Group



Click on Review and launch

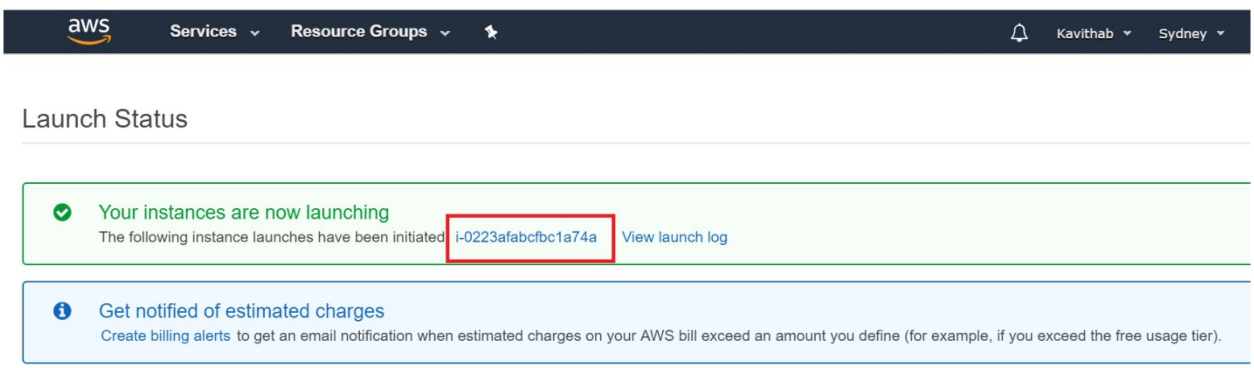


Click on Lanuch

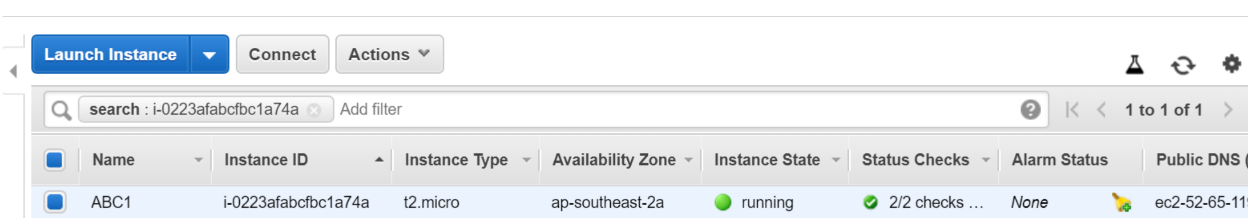


Choose the existing pair key or create the new one

Then click on Launch Instances

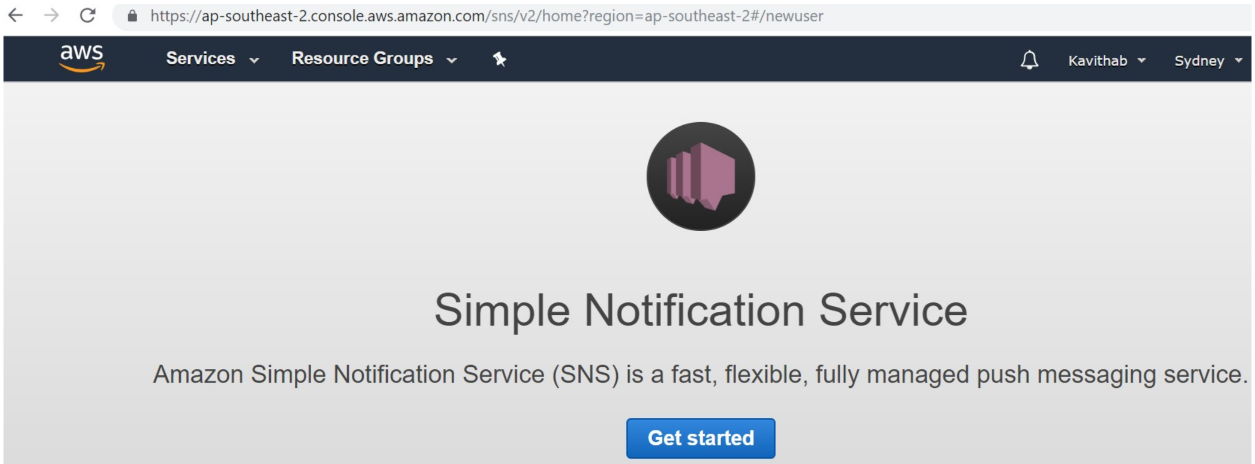


Click on the id which is marked in the above screen

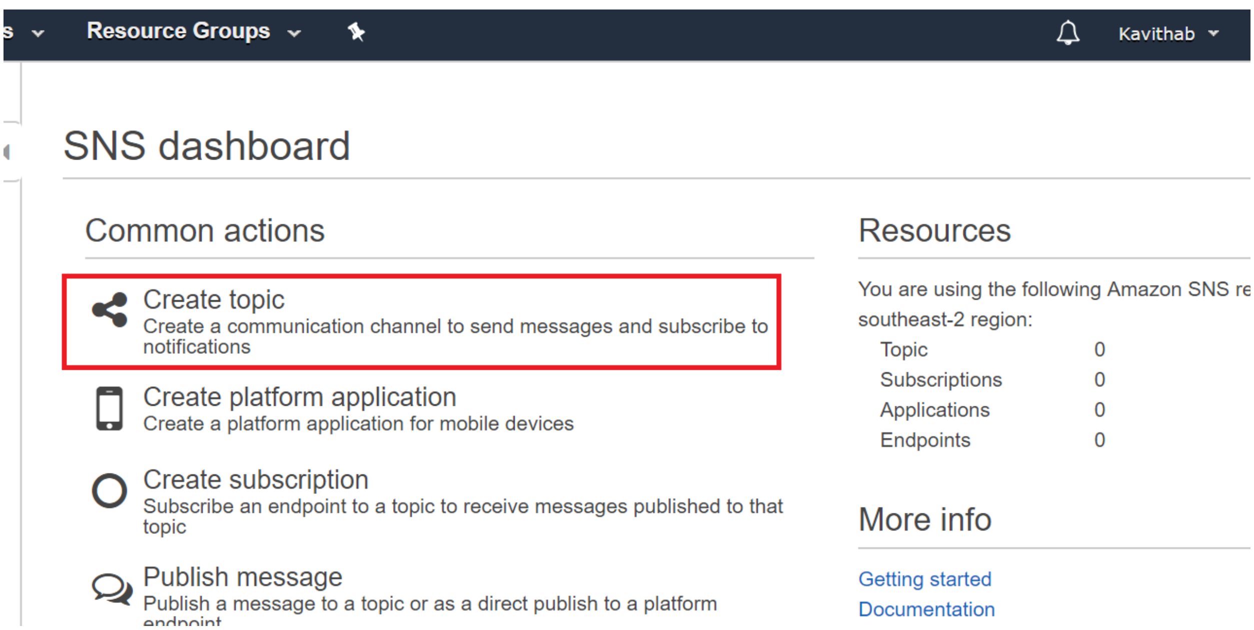


Your instance is Launched now.

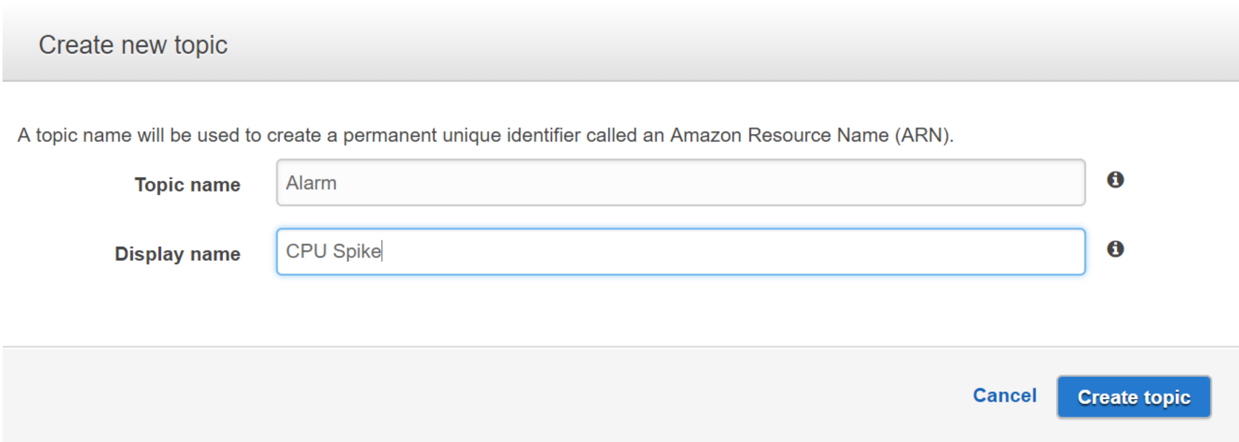
From the services select Simple Notification Services



Click on Get Started



Click on Create Topic



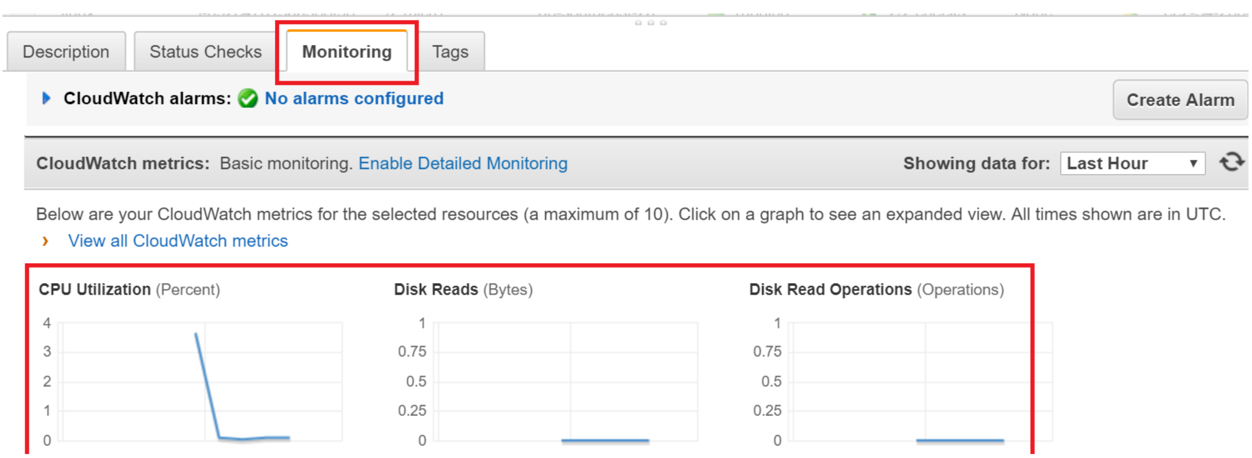
Enter the Topic name

Display name (Incident topic) In the above screen shot we are generating alarm for CPU Utilization, set display name as CPU Spike

Then click on Create Topic

Go to Instance

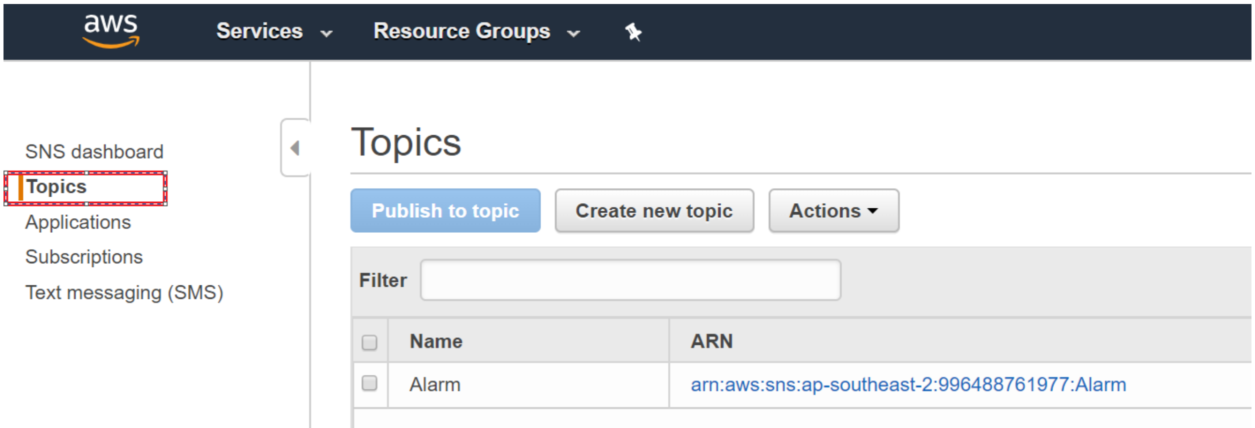
Click on Monitoring

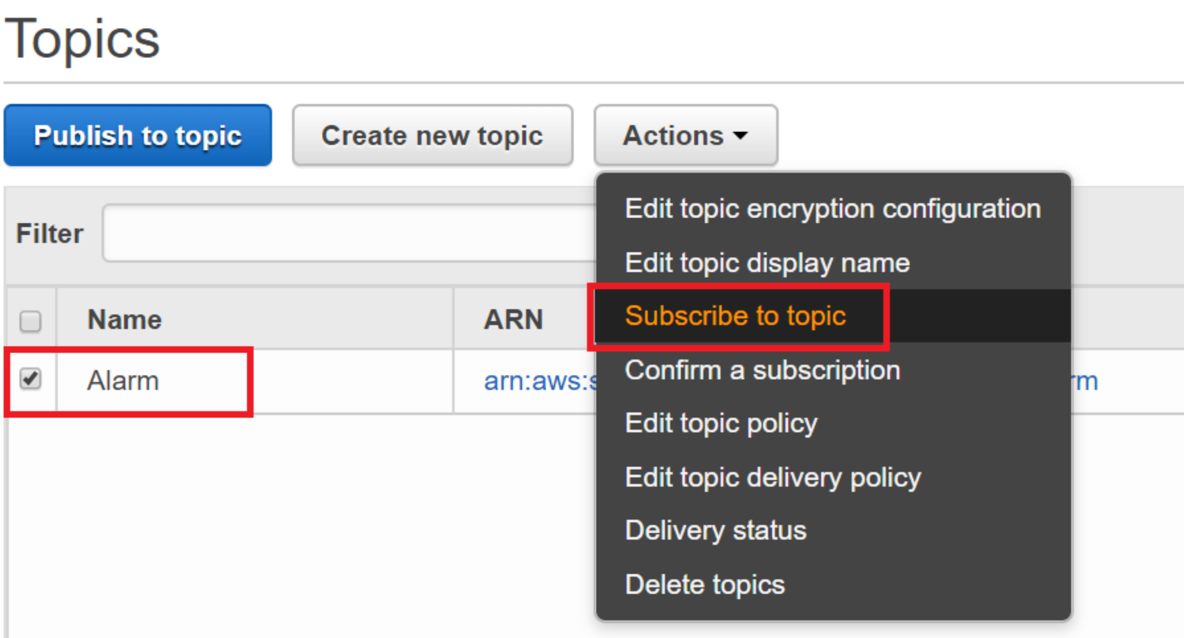


We can view the Cpu utilization details

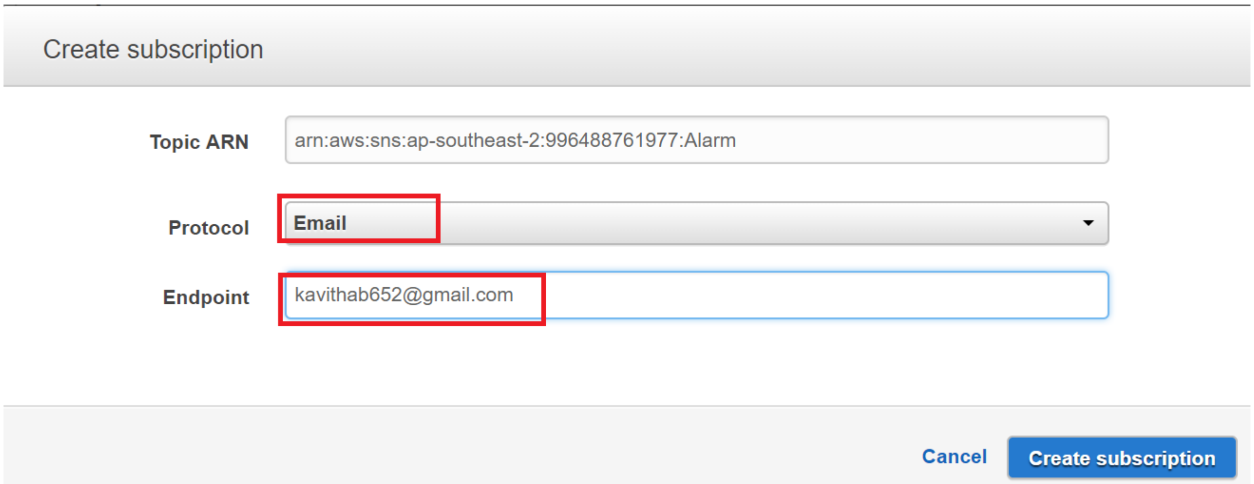
Then go to SNS (simple notification services)

Click on Topics





Select your topic and from Actions select Subscribe topic

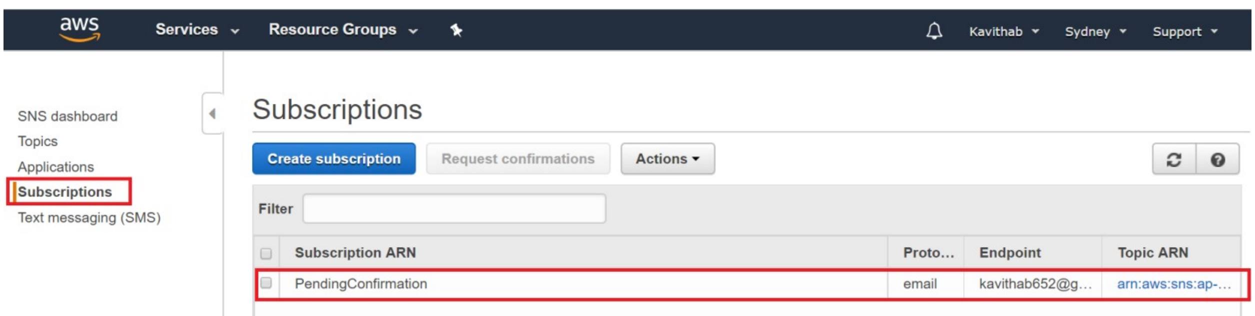


Protocol ---Select your Services as Email

Endpoint --- Mail id

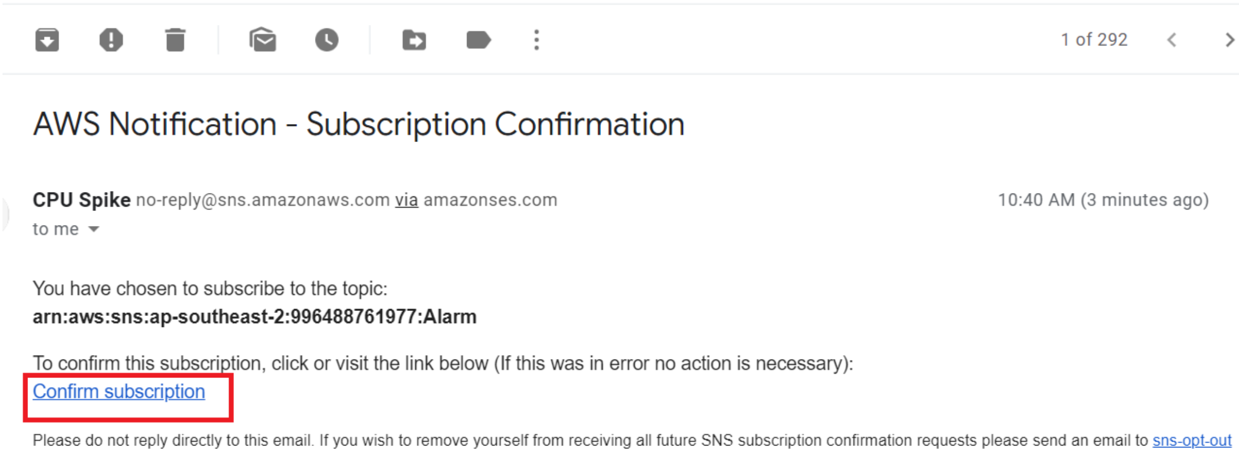
Click on Create Subscription

Click on subscription

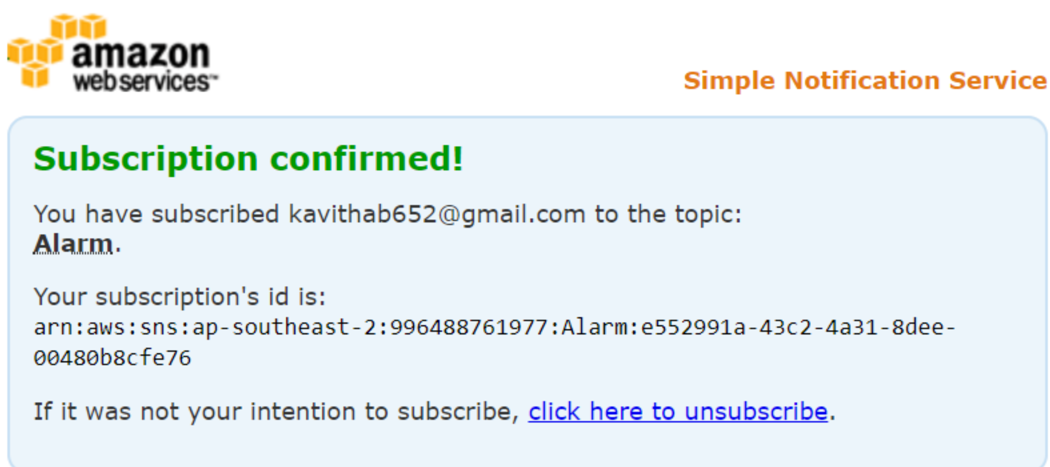


You will get a mail for confirmation for the alerts.

Go to your mail box



Open the notification mail and click on Confirm Subscription

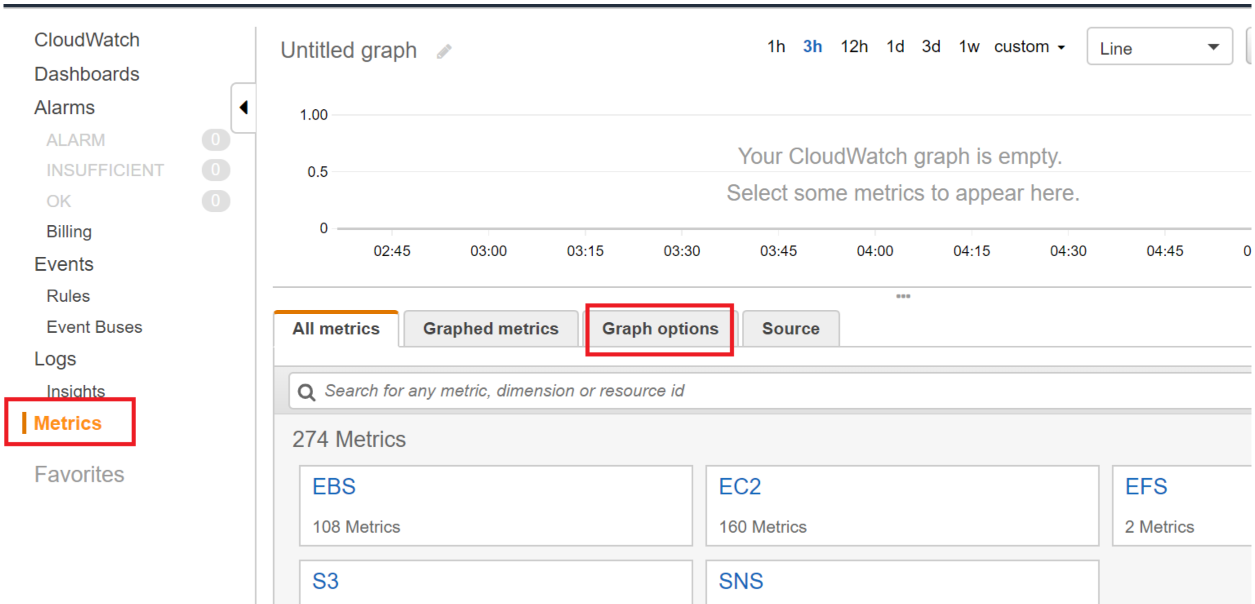


Then go to SNS and click on Refresh button

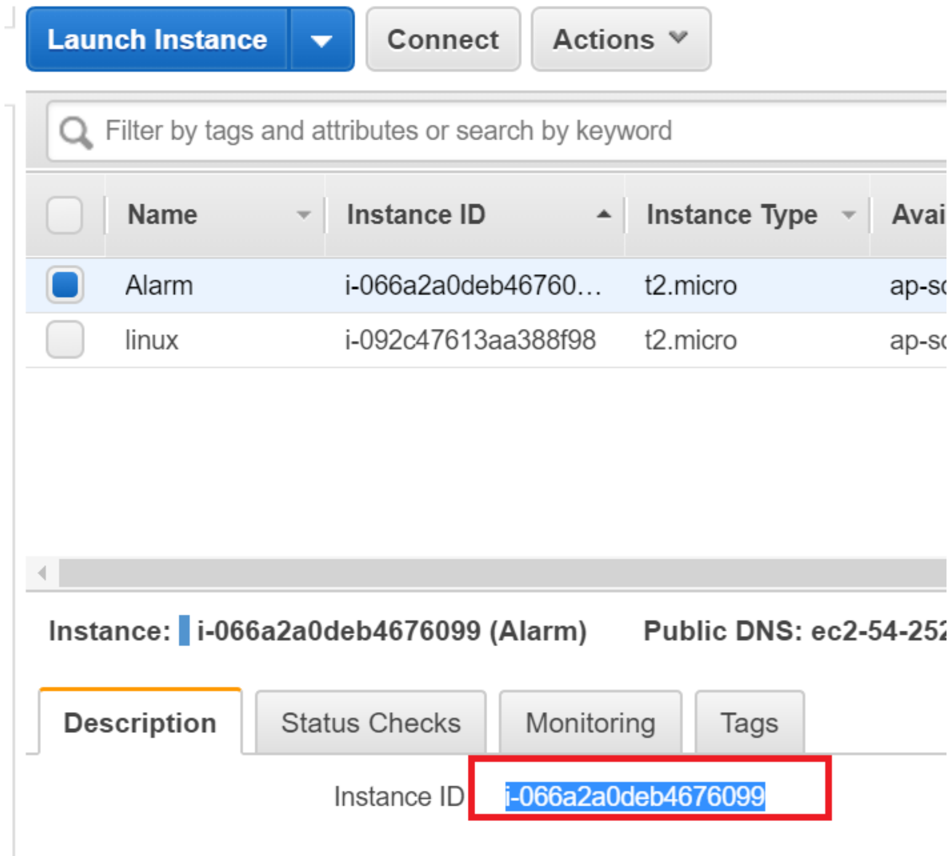
Then go to services

Select Cloud Watch

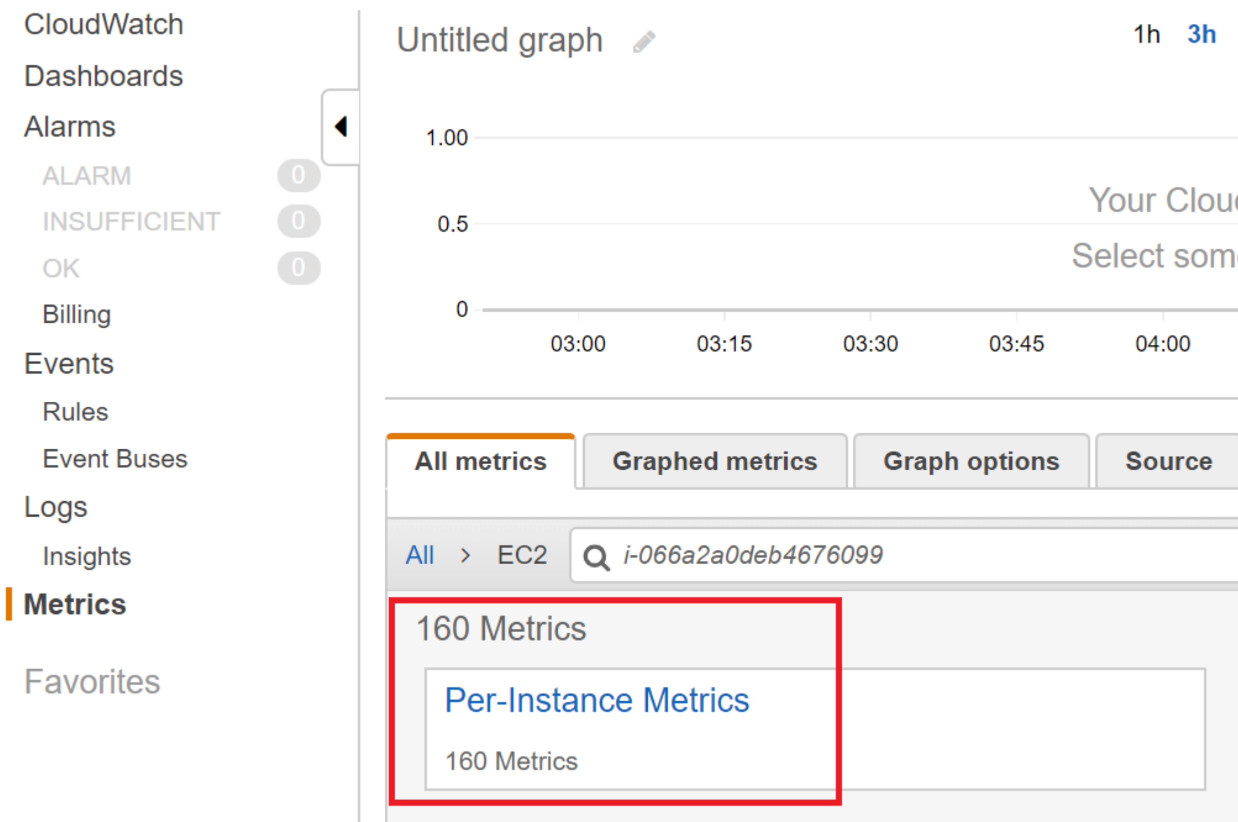
Then click on Metrics



Copy the instance ID from Instance

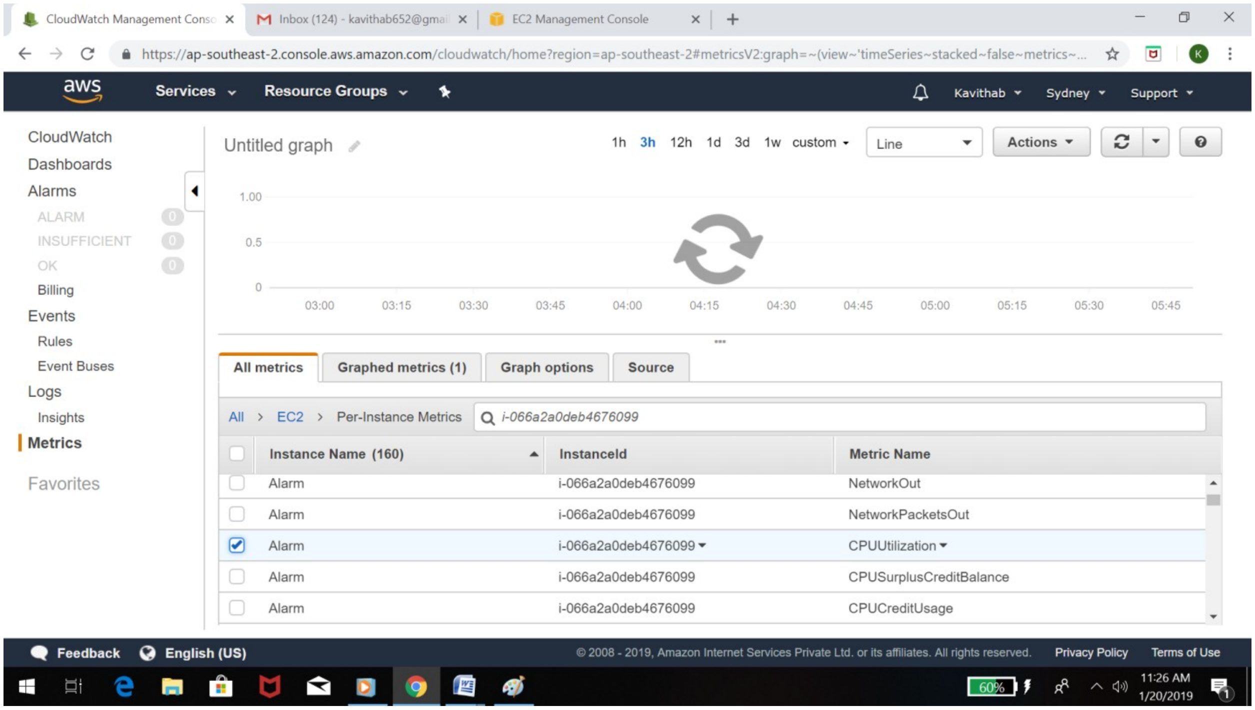


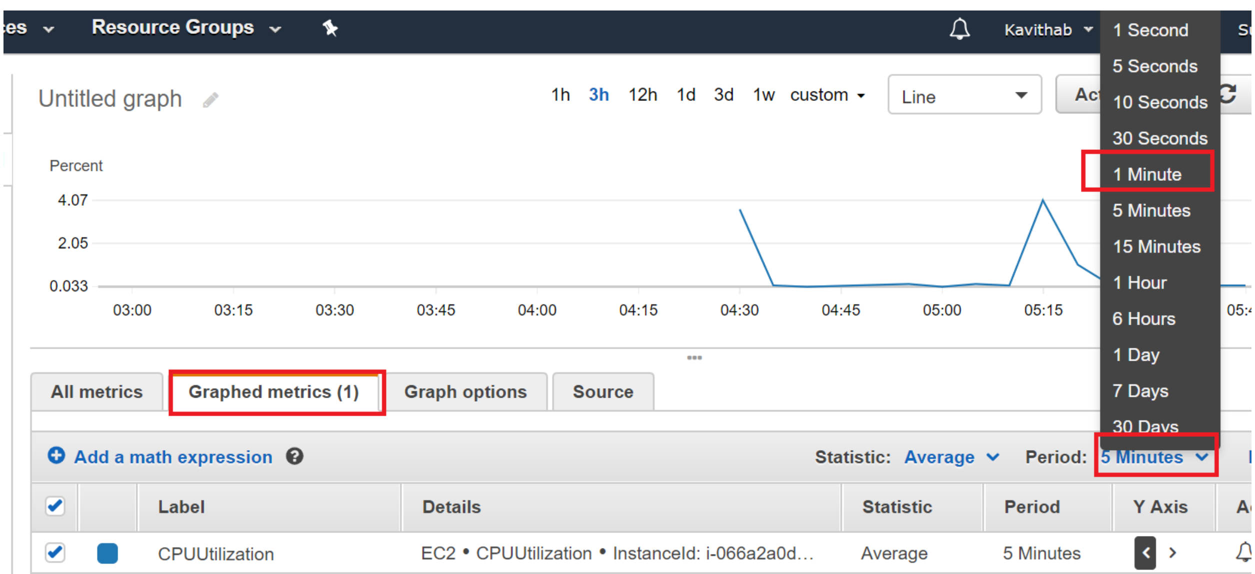
Then go to Clouc watch -→Metrics -→ EC2---→ Pre instance Metrics



Then copy and paste the Instance id

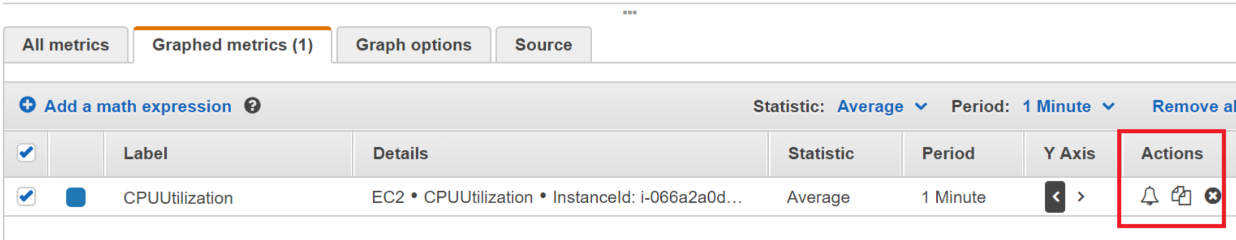
Then select the Alarm of CPU Utilization from the given list below



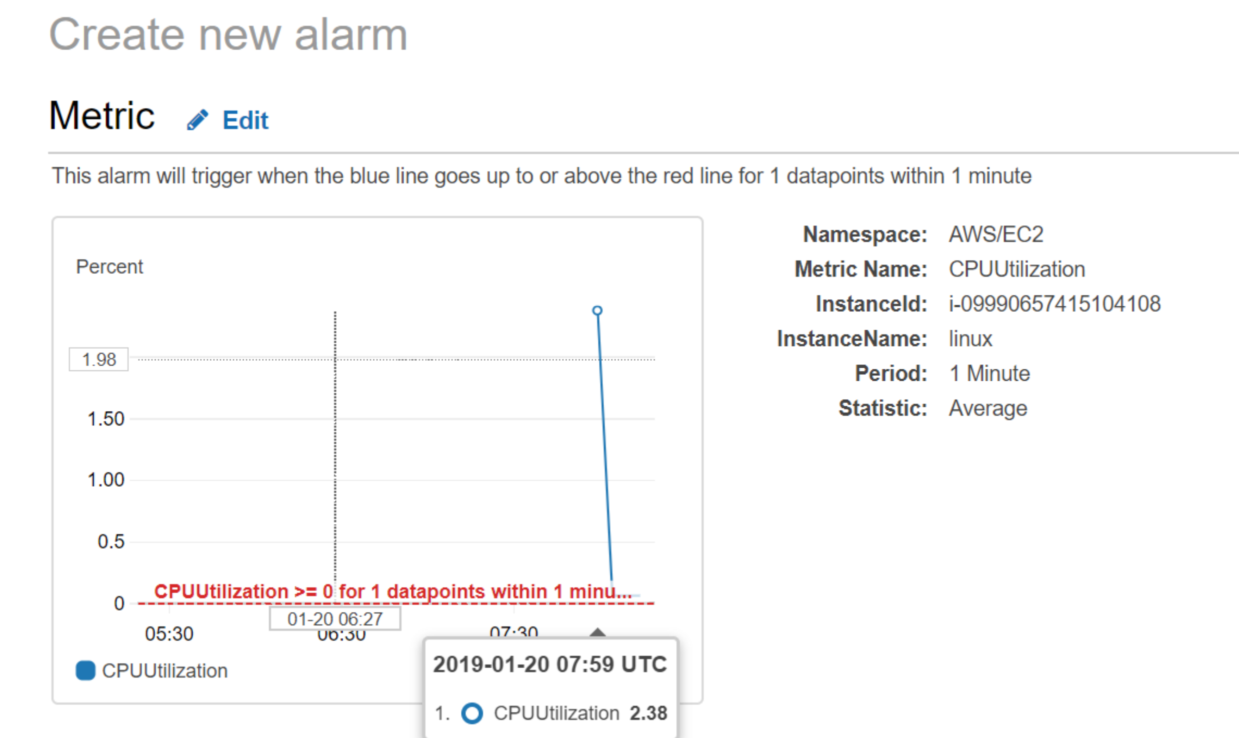


Select Graphed Metrics -→ Drop down the Time Period - we can set the time as per our requirement

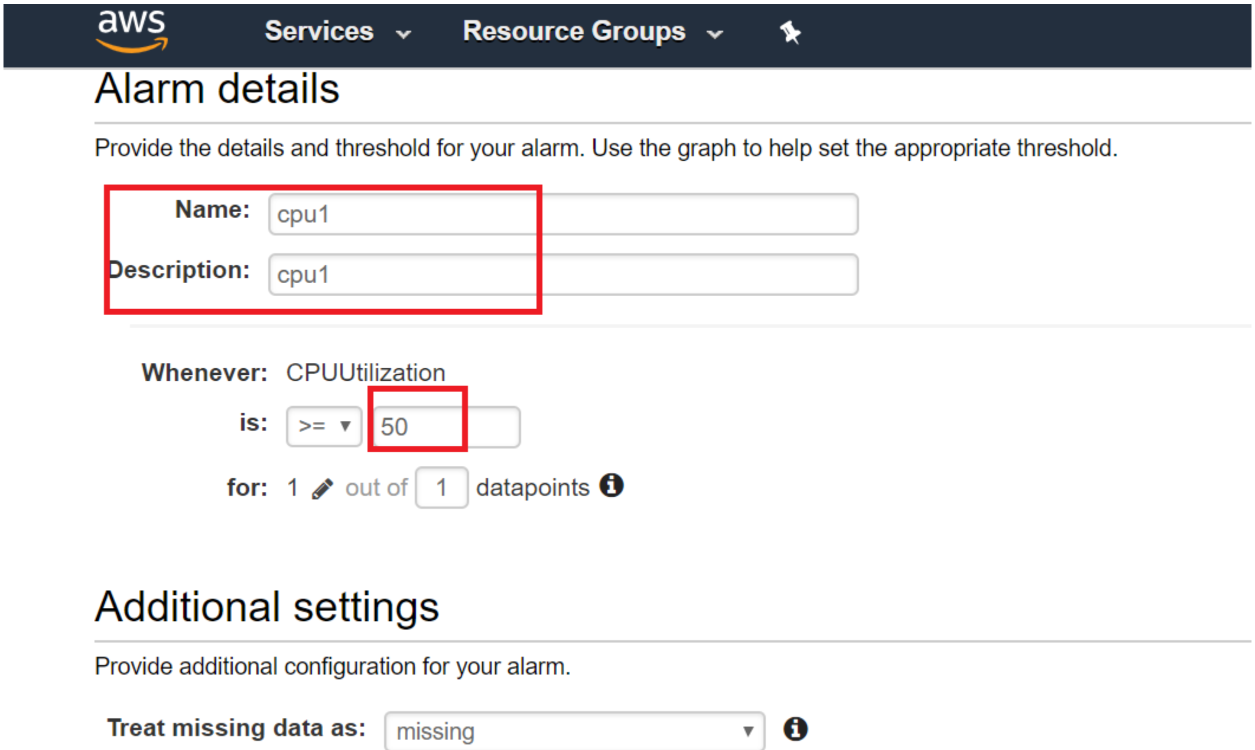
In the above screen set for 1 minute



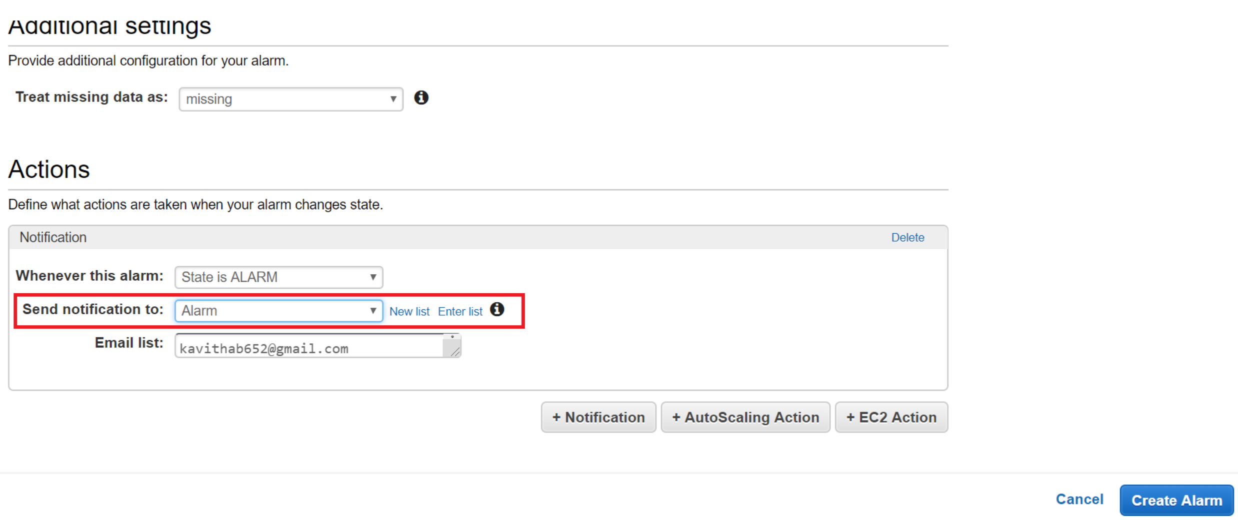
Then Click on Y Axis and then Click Bell Icon



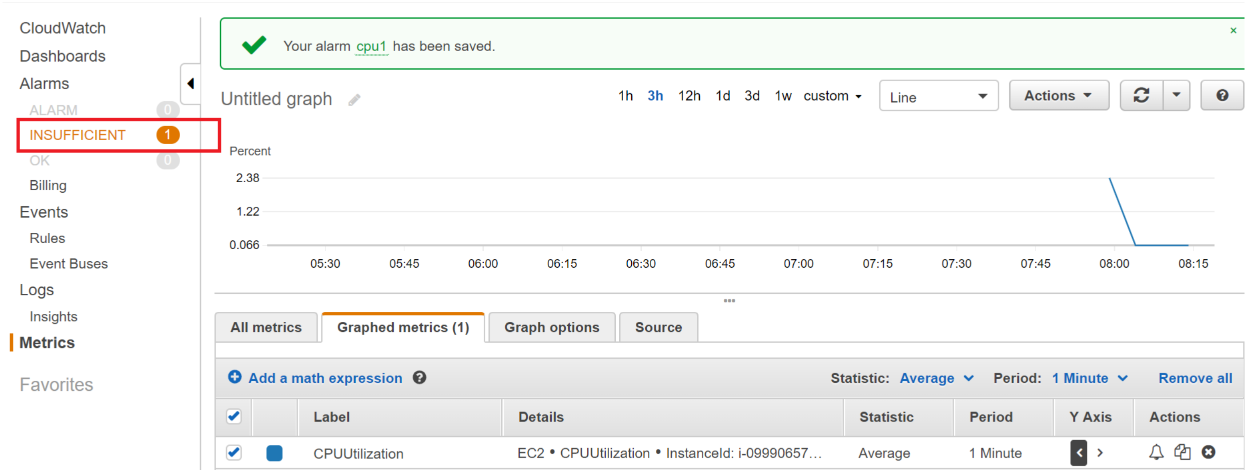
Then Enter the name and description for Alarm as mentioned in the below screenshots



Select the Send notification to as Alarm from the drop down list

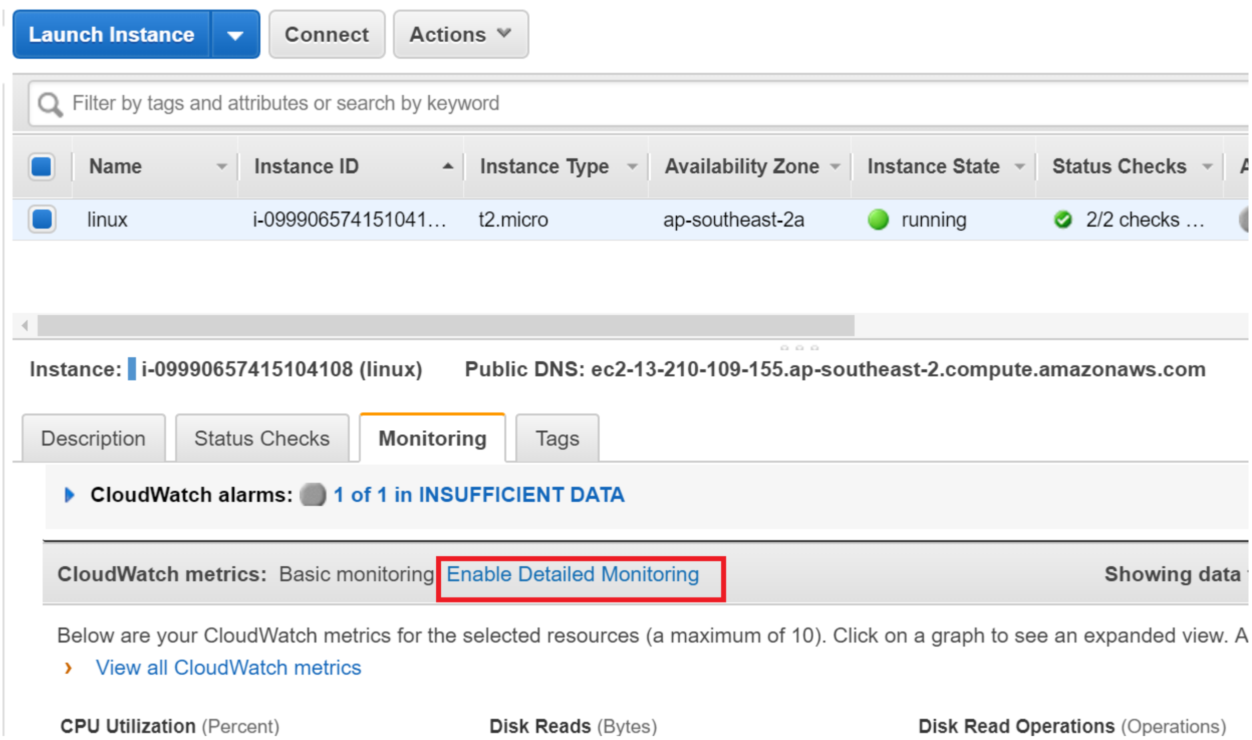


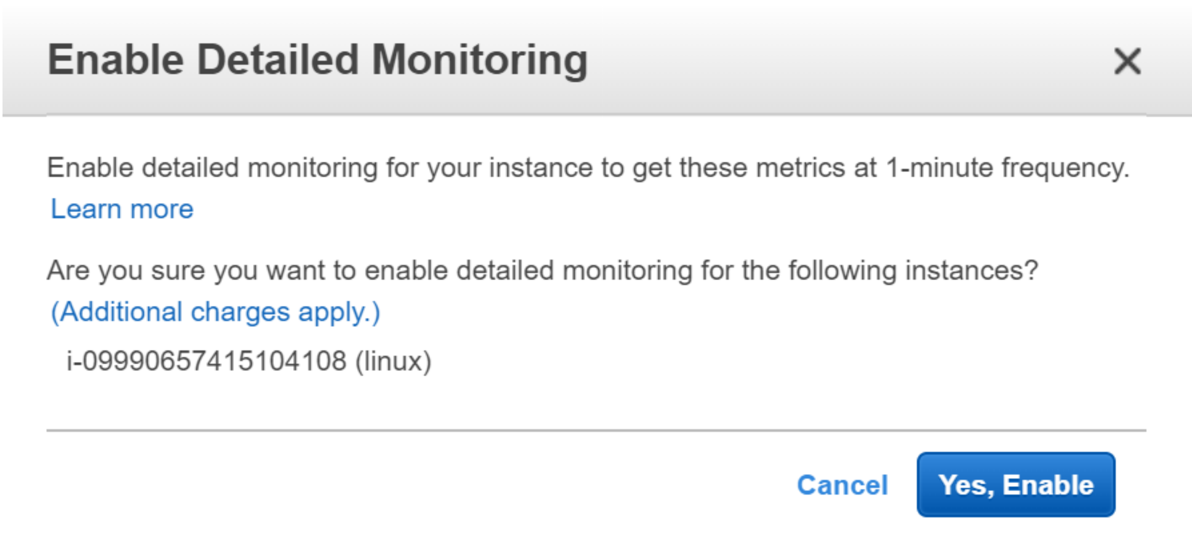
Then Click on Create Alarm



Now we have to enable the detailed Monitoring

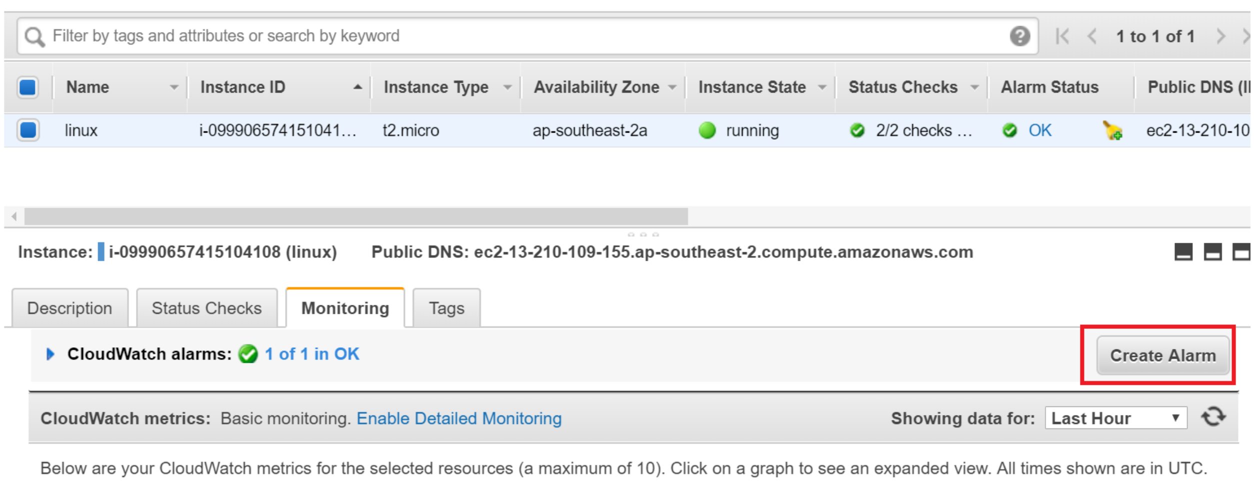
Then go to Instance Right click on the Enable Detailed Monitoring to enable the

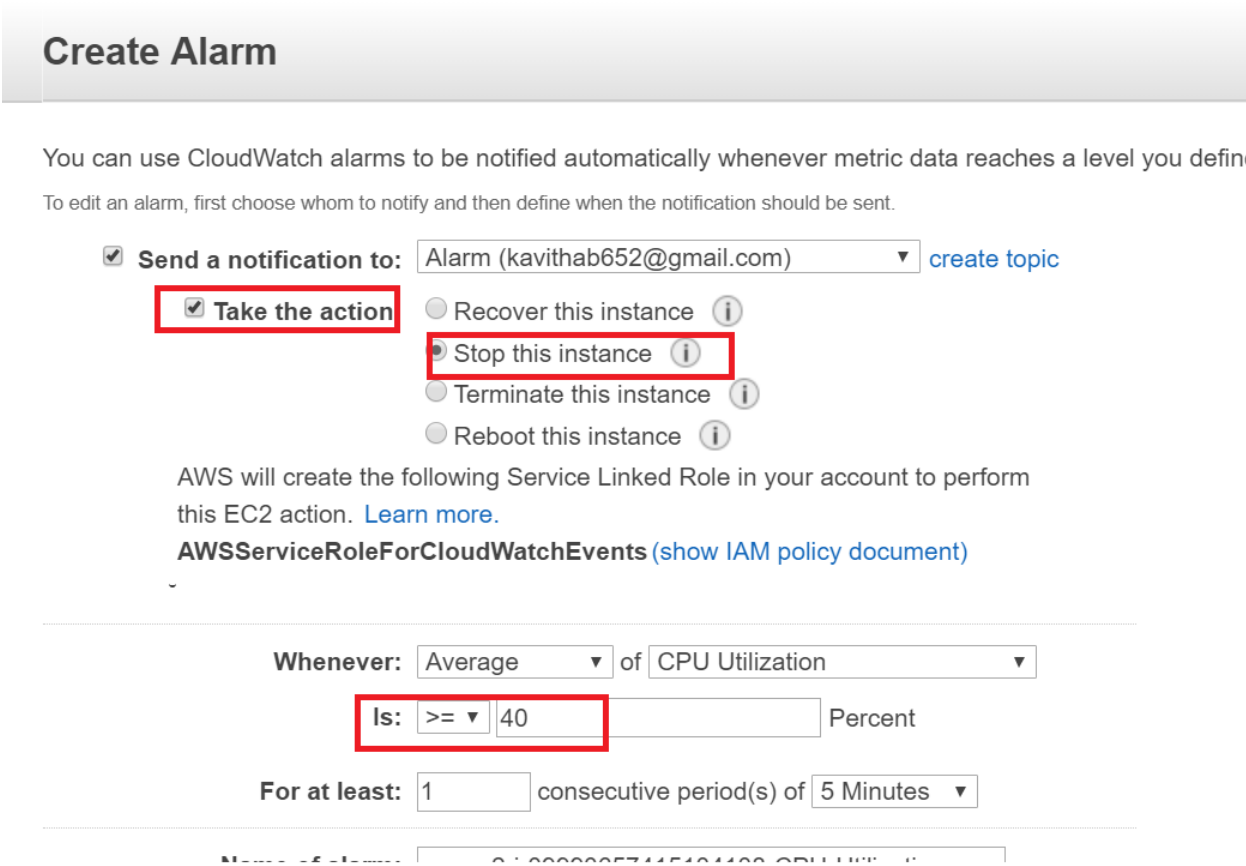




Then Click on Yes Enable

Then go to Instance ---→ Monitoring ------click on Create Alarm

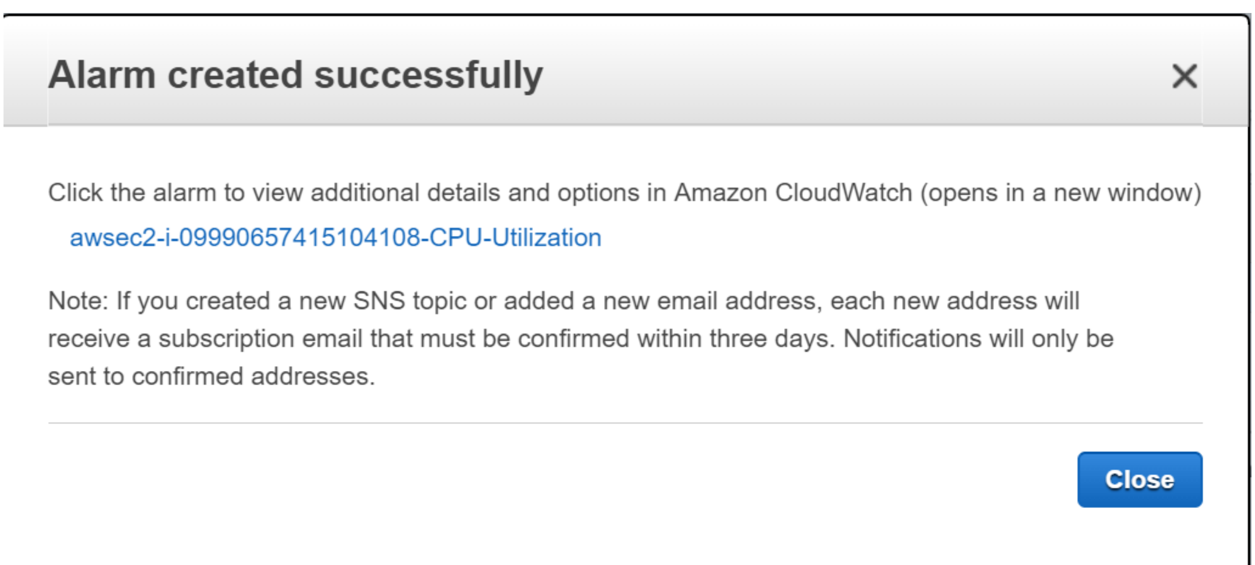




Select the Take action Check box- Choose the option as per your requirement

Here we selected as Stop this instance

Set the threshold value



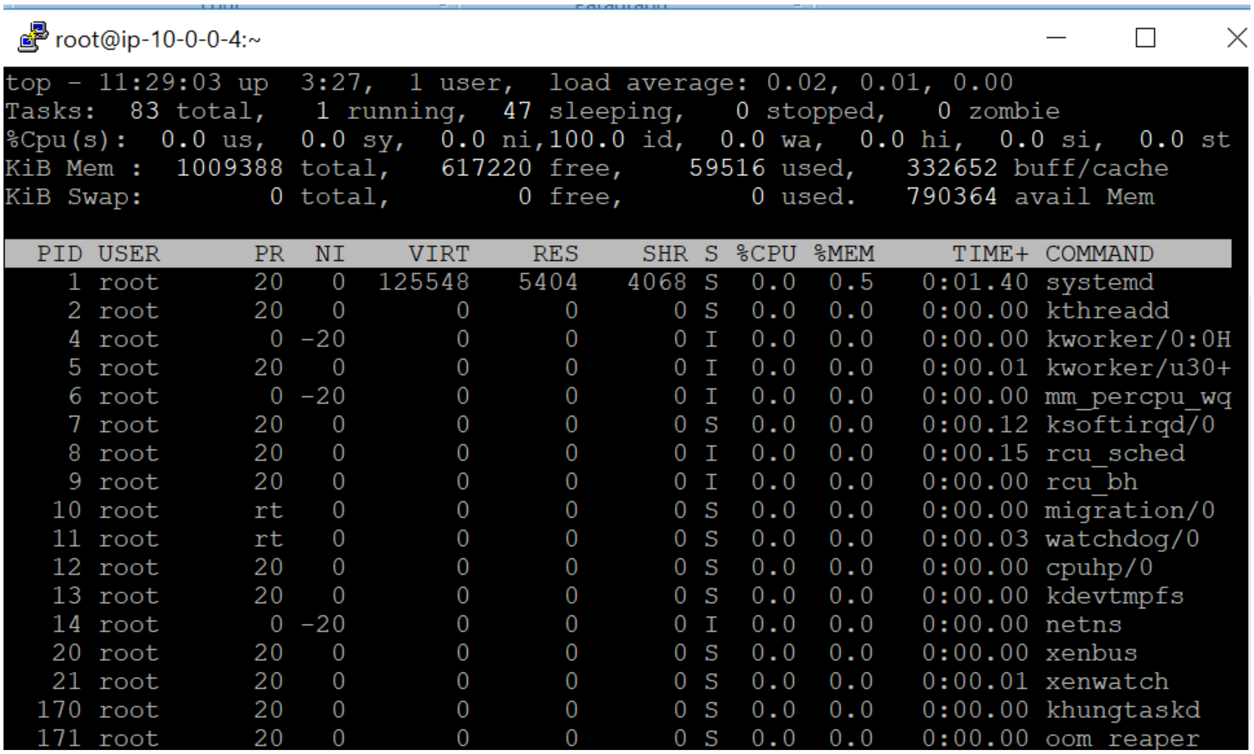
Go to OS level (Putty)

Login with User id - ec2-user

Execute below commands

 # sudo –i

 # top



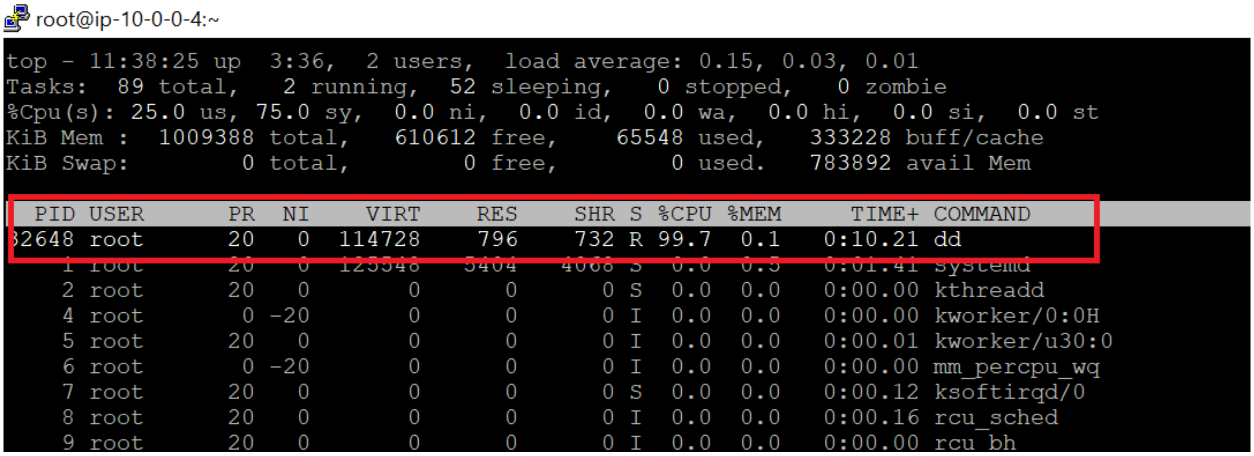

```
top - 11:29:03 up  3:27,  1 user,  load average: 0.02, 0.01, 0.00
Tasks:  83 total,   1 running,  47 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
KiB Mem :  1009388 total,   617220 free,    59516 used,   332652 buff/cache
KiB Swap:        0 total,        0 free,        0 used.   790364 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S %CPU %MEM     TIME+ COMMAND
    1 root      20   0  125548   5404   4068 S  0.0  0.5   0:01.40 systemd
    2 root      20   0       0      0      0 S  0.0  0.0   0:00.00 kthreadd
    4 root       0 -20       0      0      0 I  0.0  0.0   0:00.00 kworker/0:0H
    5 root      20   0       0      0      0 I  0.0  0.0   0:00.01 kworker/u30+
    6 root       0 -20       0      0      0 I  0.0  0.0   0:00.00 mm_percpu_wq
    7 root      20   0       0      0      0 S  0.0  0.0   0:00.12 ksoftirqd/0
    8 root      20   0       0      0      0 I  0.0  0.0   0:00.15 rcu_sched
    9 root      20   0       0      0      0 I  0.0  0.0   0:00.00 rcu_bh
   10 root      rt   0       0      0      0 S  0.0  0.0   0:00.00 migration/0
   11 root      rt   0       0      0      0 S  0.0  0.0   0:00.03 watchdog/0
   12 root      20   0       0      0      0 S  0.0  0.0   0:00.00 cpuhp/0
   13 root      20   0       0      0      0 S  0.0  0.0   0:00.00 kdevtmpfs
   14 root       0 -20       0      0      0 I  0.0  0.0   0:00.00 netns
   20 root      20   0       0      0      0 S  0.0  0.0   0:00.00 xenbus
   21 root      20   0       0      0      0 S  0.0  0.0   0:00.01 xenwatch
  170 root      20   0       0      0      0 S  0.0  0.0   0:00.00 khungtaskd
  171 root      20   0       0      0      0 S  0.0  0.0   0:00.00 oom_reaper
```

In the above screen Utilization of CPU is 0%

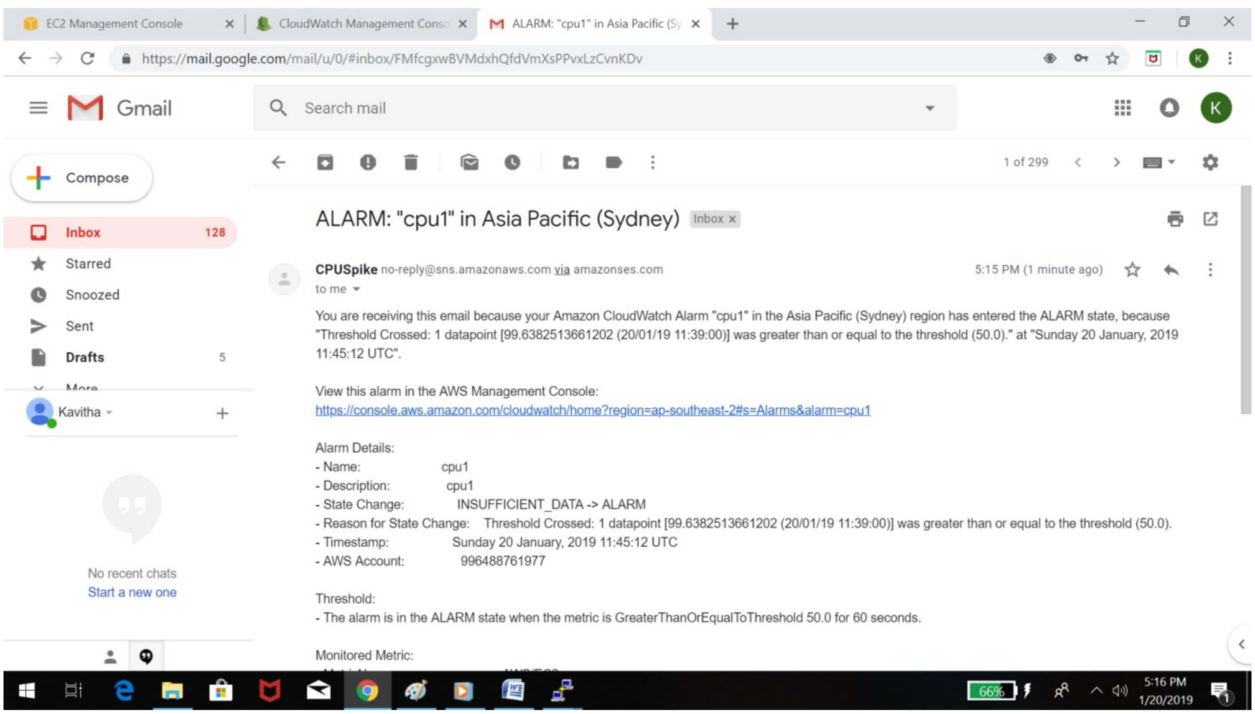
To increase the cpu utilization execute the below command to generate alarm

 # dd if=/dev/zero of=/dev/null



In the above screen CPU Utilization increased to 99.7%

We got the Alarm (alert) through mail



Once the Alarm generated and the instance is stopped.